$$
\begin{aligned}
&C_{qq}^{(3)}\mathrm{i1\_i2\_i3\_i4\_} \to \\
&\frac{1}{16\pi^2}\ (-\frac{1}{288}\ \frac{1}{m_\Phi^2}\ (s_\gamma^2\ \overline{y_d}^{i4p}\ (y_d^{i3p}\ (18\ c_\gamma^2\ \overline{y_u}^{i2r}\ y_u^{i1r} + g_2^2\ \delta_{i1i2}) + 4\ g_3^2\ y_d^{i1p}\ \delta_{i2i3}) + s_\gamma^2 \\
&\qquad \overline{y_d}^{i2p}\ (4\ g_3^2\ y_d^{i3p}\ \delta_{i1i4} + y_d^{i1p}\ (18\ c_\gamma^2\ (\overline{y_d}^{i4r}\ y_d^{i3r} + \overline{y_u}^{i4r}\ y_u^{i3r}) + g_2^2\ \delta_{i3i4})) + \\
&\qquad c_\gamma^2\ (\overline{y_u}^{i4p}\ (g_2^2\ y_u^{i3p}\ \delta_{i1i2} + 4\ g_3^2\ y_u^{i1p}\ \delta_{i2i3}) + \\
&\qquad \overline{y_u}^{i2p}\ (4\ g_3^2\ y_u^{i3p}\ \delta_{i1i4} + y_u^{i1p}\ (18\ s_\gamma^2\ \overline{y_u}^{i4r}\ y_u^{i3r} + g_2^2\ \delta_{i3i4})))) + \\
&\frac{1}{18}\ g_2^4\ LF_{3,0}[m_2]\ \delta_{i1i2}\ \delta_{i3i4} + \frac{1}{12}\ g_2^4\ LF_{4,-1}[m_2]\ \delta_{i1i2}\ \delta_{i3i4} - \frac{4}{45}\ g_2^4\ LF_{5,-2}[m_2]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{1}{12}\ g_3^4\ LF_{3,0}[m_3]\ \delta_{i1i4}\ \delta_{i2i3} + \frac{1}{8}\ g_3^4\ LF_{4,-1}[m_3]\ \delta_{i1i4}\ \delta_{i2i3} - \\
&\frac{2}{15}\ g_3^4\ LF_{5,-2}[m_3]\ \delta_{i1i4}\ \delta_{i2i3} + \frac{1}{36}\ \sum_p g_3^4\ LF_{3,0}[m_{\tilde{d}}^p]\ \delta_{i1i4}\ \delta_{i2i3} - \\
&\frac{5}{96}\ \sum_p g_3^4\ LF_{4,-1}[m_{\tilde{d}}^p]\ \delta_{i1i4}\ \delta_{i2i3} + \\
&\frac{1}{45}\ \sum_p g_3^4\ LF_{5,-2}[m_{\tilde{d}}^p]\ \delta_{i1i4}\ \delta_{i2i3} + \frac{1}{36}\ \sum_p g_2^4\ LF_{3,0}[m_{\tilde{l}}^p]\ \delta_{i1i2}\ \delta_{i3i4} - \\
&\frac{5}{96}\ \sum_p g_2^4\ LF_{4,-1}[m_{\tilde{l}}^p]\ \delta_{i1i2}\ \delta_{i3i4} + \frac{1}{45}\ \sum_p g_2^4\ LF_{5,-2}[m_{\tilde{l}}^p]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{1}{36}\ \sum_p\ (2\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 3\ g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,0}[m_{\tilde{q}}^p] - \\
&\frac{5}{96}\ \sum_p\ (2\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 3\ g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,-1}[m_{\tilde{q}}^p] + \\
&\frac{1}{45}\ \sum_p\ (2\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 3\ g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{5,-2}[m_{\tilde{q}}^p] + \frac{1}{36}\ \sum_p g_3^4\ LF_{3,0}[m_{\tilde{u}}^p]\ \delta_{i1i4}\ \delta_{i2i3} - \\
&\frac{5}{96}\ \sum_p g_3^4\ LF_{4,-1}[m_{\tilde{u}}^p]\ \delta_{i1i4}\ \delta_{i2i3} + \frac{1}{45}\ \sum_p g_3^4\ LF_{5,-2}[m_{\tilde{u}}^p]\ \delta_{i1i4}\ \delta_{i2i3} + \\
&\frac{1}{24}\ (s_\gamma^2\ \overline{y_d}^{i2p}\ (3\ c_\gamma^2\ y_d^{i1p}\ (\overline{y_d}^{i4r}\ y_d^{i3r} - \overline{y_u}^{i4r}\ y_u^{i3r}) - g_3^2\ y_d^{i3p}\ \delta_{i1i4}) - \\
&\qquad s_\gamma^2\ \overline{y_d}^{i4p}\ (3\ c_\gamma^2\ y_d^{i3p}\ \overline{y_u}^{i2r}\ y_u^{i1r} + g_3^2\ y_d^{i1p}\ \delta_{i2i3}) + \\
&\qquad c_\gamma^2\ (\overline{y_u}^{i2p}\ (3\ s_\gamma^2\ \overline{y_u}^{i4r}\ y_u^{i1p}\ y_u^{i3r} - g_3^2\ y_u^{i3p}\ \delta_{i1i4}) - g_3^2\ \overline{y_u}^{i4p}\ y_u^{i1p}\ \delta_{i2i3})) \\
&\quad LF_{1,2}[m_\Phi] + \frac{1}{48}\ (s_\gamma^2\ \overline{y_d}^{i4p}\ y_d^{i3p}\ (3\ c_\gamma^2\ \overline{y_u}^{i2r}\ y_u^{i1r} - g_2^2\ \delta_{i1i2}) + \\
&\qquad s_\gamma^2\ \overline{y_d}^{i2p}\ y_d^{i1p}\ (3\ s_\gamma^2\ \overline{y_d}^{i4r}\ y_d^{i3r} + 3\ c_\gamma^2\ \overline{y_u}^{i4r}\ y_u^{i3r} - g_2^2\ \delta_{i3i4}) + \\
&\qquad c_\gamma^2\ (-g_2^2\ \overline{y_u}^{i4p}\ y_u^{i3p}\ \delta_{i1i2} + \overline{y_u}^{i2p}\ y_u^{i1p}\ (3\ c_\gamma^2\ \overline{y_u}^{i4r}\ y_u^{i3r} - g_2^2\ \delta_{i3i4})))\ LF_{2,1}[m_\Phi] + \\
&\frac{1}{144}\ g_2^2\ (3\ (s_\gamma^2\ \overline{y_d}^{i4p}\ y_d^{i3p} + c_\gamma^2\ \overline{y_u}^{i4p}\ y_u^{i3p})\ \delta_{i1i2} + \\
&\qquad (3\ s_\gamma^2\ \overline{y_d}^{i2p}\ y_d^{i1p} + 3\ c_\gamma^2\ \overline{y_u}^{i2p}\ y_u^{i1p} + 4\ g_2^2\ \delta_{i1i2})\ \delta_{i3i4})\ LF_{3,0}[m_\Phi] - \\
&\frac{5}{96}\ g_2^4\ LF_{4,-1}[m_\Phi]\ \delta_{i1i2}\ \delta_{i3i4} + \frac{1}{45}\ g_2^4\ LF_{5,-2}[m_\Phi]\ \delta_{i1i2}\ \delta_{i3i4} + \frac{1}{36}\ g_2^4\ LF_{3,0}[\tilde{\mu}]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{1}{24}\ g_2^4\ LF_{4,-1}[\tilde{\mu}]\ \delta_{i1i2}\ \delta_{i3i4} - \frac{2}{45}\ g_2^4\ LF_{5,-2}[\tilde{\mu}]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_1,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_1,\ m_{\tilde{q}}^{i1}] - \\
&\frac{1}{864}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_1,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_1,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_1,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_1,\ m_{\tilde{q}}^{i2}] - \\
&\frac{1}{864}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_1,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_1,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_1,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_1,\ m_{\tilde{q}}^{i3}] - \\
&\frac{1}{864}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_1,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_1,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_1,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_1,\ m_{\tilde{q}}^{i4}] - \\
&\frac{1}{864}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_1,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_1,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_2,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_2,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{96}\ (-3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - 5\ g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_2,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{64}\ (g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_2,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_2,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_2,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{96}\ (-3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - 5\ g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_2,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{64}\ (g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_2,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_2,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_2,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{96}\ (-3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - 5\ g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_2,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{64}\ (g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_2,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_2,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_2,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{96}\ (-3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - 5\ g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_2,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{64}\ (g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_2,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_3,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_3,\ m_{\tilde{q}}^{i1}] - \\
&\frac{1}{288}\ g_3^2\ (25\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + 16\ g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_3,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{36}\ (g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_3,\ m_{\tilde{q}}^{i1}] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_3,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_3,\ m_{\tilde{q}}^{i2}] - \\
&\frac{1}{288}\ g_3^2\ (25\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + 16\ g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_3,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{36}\ (g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_3,\ m_{\tilde{q}}^{i2}] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_3,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_3,\ m_{\tilde{q}}^{i3}] - \\
&\frac{1}{288}\ g_3^2\ (25\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + 16\ g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_3,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{36}\ (g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_3,\ m_{\tilde{q}}^{i3}] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_3,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,2,-1}[m_3,\ m_{\tilde{q}}^{i4}] - \\
&\frac{1}{288}\ g_3^2\ (25\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + 16\ g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_3,\ m_{\tilde{q}}^{i4}] + \\
&\frac{1}{36}\ (g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{4,1,-2}[m_3,\ m_{\tilde{q}}^{i4}] - \\
&\frac{1}{24}\ g_3^2\ (\overline{y_d}^{i2p}\ y_d^{i3p}\ \delta_{i1i4} + \overline{y_d}^{i4p}\ y_d^{i1p}\ \delta_{i2i3})\ LF_{2,1,0}[m_{\tilde{d}}^p,\ \tilde{\mu}] + \\
&\frac{1}{48}\ g_3^2\ (\overline{y_d}^{i2p}\ y_d^{i3p}\ \delta_{i1i4} + \overline{y_d}^{i4p}\ y_d^{i1p}\ \delta_{i2i3})\ LF_{2,2,-1}[m_{\tilde{d}}^p,\ \tilde{\mu}] + \\
&\frac{1}{48}\ g_3^2\ (\overline{y_d}^{i2p}\ y_d^{i3p}\ \delta_{i1i4} + \overline{y_d}^{i4p}\ y_d^{i1p}\ \delta_{i2i3})\ LF_{3,1,-1}[m_{\tilde{d}}^p,\ \tilde{\mu}] - \\
&\frac{1}{864}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i1},\ m_1] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i1},\ m_1] + \\
&\frac{1}{96}\ (-3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i1},\ m_2] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i1},\ m_2] + \\
&\frac{1}{144}\ (g_3^4\ \delta_{i1i4}\ \delta_{i2i3} - 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i1},\ m_3] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i1},\ m_3] - \\
&\frac{1}{864}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i2},\ m_1] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i2},\ m_1] + \\
&\frac{1}{96}\ (-3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i2},\ m_2] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i2},\ m_2] + \\
&\frac{1}{144}\ (g_3^4\ \delta_{i1i4}\ \delta_{i2i3} - 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i2},\ m_3] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i2},\ m_3] - \\
&\frac{1}{864}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i3},\ m_1] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i3},\ m_1] + \\
&\frac{1}{96}\ (-3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i3},\ m_2] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i3},\ m_2] + \\
&\frac{1}{144}\ (g_3^4\ \delta_{i1i4}\ \delta_{i2i3} - 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i3},\ m_3] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i3},\ m_3] - \\
&\frac{1}{864}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i4},\ m_1] + \\
&\frac{1}{1728}\ g_1^2\ (g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i4},\ m_1] + \\
&\frac{1}{96}\ (-3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} + g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i4},\ m_2] + \\
&\frac{1}{192}\ (3\ g_2^2\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3} - g_2^4\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i4},\ m_2] + \\
&\frac{1}{144}\ (g_3^4\ \delta_{i1i4}\ \delta_{i2i3} - 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{2,1,0}[m_{\tilde{q}}^{i4},\ m_3] + \\
&\frac{1}{288}\ (-g_3^4\ \delta_{i1i4}\ \delta_{i2i3} + 8\ g_2^2\ g_3^2\ \delta_{i1i2}\ \delta_{i3i4})\ LF_{3,1,-1}[m_{\tilde{q}}^{i4},\ m_3] - \\
&\frac{1}{24}\ g_3^2\ (\overline{y_u}^{i2p}\ y_u^{i3p}\ \delta_{i1i4} + \overline{y_u}^{i4p}\ y_u^{i1p}\ \delta_{i2i3})\ LF_{2,1,0}[m_{\tilde{u}}^p,\ \tilde{\mu}] + \\
&\frac{1}{48}\ g_3^2\ (\overline{y_u}^{i2p}\ y_u^{i3p}\ \delta_{i1i4} + \overline{y_u}^{i4p}\ y_u^{i1p}\ \delta_{i2i3})\ LF_{2,2,-1}[m_{\tilde{u}}^p,\ \tilde{\mu}] + \\
&\frac{1}{48}\ g_3^2\ (\overline{y_u}^{i2p}\ y_u^{i3p}\ \delta_{i1i4} + \overline{y_u}^{i4p}\ y_u^{i1p}\ \delta_{i2i3})\ LF_{3,1,-1}[m_{\tilde{u}}^p,\ \tilde{\mu}] + \\
&\frac{1}{48}\ g_3^2\ (\overline{y_d}^{i2p}\ y_d^{i3p}\ \delta_{i1i4} + \overline{y_d}^{i4p}\ y_d^{i1p}\ \delta_{i2i3})\ LF_{2,1,0}[\tilde{\mu},\ m_{\tilde{d}}^p] + \\
&\frac{1}{48}\ (-\overline{y_d}^{i4p}\ (3\ g_2^2\ y_d^{i3p}\ \delta_{i1i2} + 2\ g_3^2\ y_d^{i1p}\ \delta_{i2i3}) - \overline{y_d}^{i2p}\ (2\ g_3^2\ y_d^{i3p}\ \delta_{i1i4} + 3\ g_2^2\ y_d^{i1p}\ \delta_{i3i4})) \\
&\quad LF_{3,1,-1}[\tilde{\mu},\ m_{\tilde{d}}^p] + \\
&\frac{1}{48}\ (\overline{y_d}^{i4p}\ (g_2^2\ y_d^{i3p}\ \delta_{i1i2} + g_3^2\ y_d^{i1p}\ \delta_{i2i3}) + \overline{y_d}^{i2p}\ (g_3^2\ y_d^{i3p}\ \delta_{i1i4} + g_2^2\ y_d^{i1p}\ \delta_{i3i4})) \\
&\quad LF_{4,1,-2}[\tilde{\mu},\ m_{\tilde{d}}^p] + \frac{1}{48}\ g_3^2\ (\overline{y_u}^{i2p}\ y_u^{i3p}\ \delta_{i1i4} + \overline{y_u}^{i4p}\ y_u^{i1p}\ \delta_{i2i3})\ LF_{2,1,0}[\tilde{\mu},\ m_{\tilde{u}}^p] + \\
&\frac{1}{48}\ (-\overline{y_u}^{i4p}\ (3\ g_2^2\ y_u^{i3p}\ \delta_{i1i2} + 2\ g_3^2\ y_u^{i1p}\ \delta_{i2i3}) - \overline{y_u}^{i2p}\ (2\ g_3^2\ y_u^{i3p}\ \delta_{i1i4} + 3\ g_2^2\ y_u^{i1p}\ \delta_{i3i4})) \\
&\quad LF_{3,1,-1}[\tilde{\mu},\ m_{\tilde{u}}^p] + \\
&\frac{1}{48}\ (\overline{y_u}^{i4p}\ (g_2^2\ y_u^{i3p}\ \delta_{i1i2} + g_3^2\ y_u^{i1p}\ \delta_{i2i3}) + \overline{y_u}^{i2p}\ (g_3^2\ y_u^{i3p}\ \delta_{i1i4} + g_2^2\ y_u^{i1p}\ \delta_{i3i4})) \\
&\quad LF_{4,1,-2}[\tilde{\mu},\ m_{\tilde{u}}^p] + \frac{1}{32}\ g_2^4\ LF_{2,1,1,-1}[m_2,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4} - \\
&\frac{1}{16}\ g_2^4\ m_2^2\ LF_{2,1,1,0}[m_2,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4} + \frac{1}{32}\ g_2^4\ LF_{2,1,1,-1}[m_2,\ m_{\tilde{q}}^{i2},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4} - \\
&\frac{1}{16}\ g_2^4\ m_2^2\ LF_{2,1,1,0}[m_2,\ m_{\tilde{q}}^{i2},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{7}{96}\ g_3^4\ LF_{2,1,1,-1}[m_3,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3} - \frac{1}{24}\ g_3^4\ m_3^2\ LF_{2,1,1,0}[m_3,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3} + \\
&\frac{7}{96}\ g_3^4\ LF_{2,1,1,-1}[m_3,\ m_{\tilde{q}}^{i3},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3} - \frac{1}{24}\ g_3^4\ m_3^2\ LF_{2,1,1,0}[m_3,\ m_{\tilde{q}}^{i3},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3} - \\
&\frac{1}{8}\ \tilde{\mu}^2\ (\overline{y_d}^{i4p}\ y_d^{i3p}\ \overline{y_u}^{i2r}\ y_u^{i1r} + \overline{y_d}^{i2p}\ y_d^{i1p}\ \overline{y_u}^{i4r}\ y_u^{i3r})\ LF_{2,1,1,0}[\tilde{\mu},\ m_{\tilde{d}}^p,\ m_{\tilde{u}}^r] + \\
&\frac{1}{16}\ \overline{y_d}^{i2p}\ \overline{y_d}^{i4r}\ y_d^{i1p}\ y_d^{i3r}\ LF_{2,1,1,-1}[\tilde{\mu},\ m_{\tilde{d}}^r,\ m_{\tilde{d}}^p] + \\
&\frac{1}{16}\ \overline{y_u}^{i2p}\ \overline{y_u}^{i4r}\ y_u^{i1p}\ y_u^{i3r}\ LF_{2,1,1,-1}[\tilde{\mu},\ m_{\tilde{u}}^r,\ m_{\tilde{u}}^p] + \\
&\frac{1}{288}\ g_1^2\ g_2^2\ LF_{1,1,1,1,-1}[m_1,\ m_2,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{1}{144}\ m_1\ m_2\ g_1^2\ g_2^2\ LF_{1,1,1,1,0}[m_1,\ m_2,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{1}{288}\ g_1^2\ g_2^2\ LF_{1,1,1,1,-1}[m_1,\ m_2,\ m_{\tilde{q}}^{i2},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{1}{144}\ m_1\ m_2\ g_1^2\ g_2^2\ LF_{1,1,1,1,0}[m_1,\ m_2,\ m_{\tilde{q}}^{i2},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4} + \\
&\frac{1}{288}\ g_1^2\ g_3^2\ LF_{1,1,1,1,-1}[m_1,\ m_3,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3} + \\
&\frac{1}{144}\ m_1\ m_3\ g_1^2\ g_3^2\ LF_{1,1,1,1,0}[m_1,\ m_3,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3} + \\
&\frac{1}{288}\ g_1^2\ g_3^2\ LF_{1,1,1,1,-1}[m_1,\ m_3,\ m_{\tilde{q}}^{i3},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3} + \\
&\frac{1}{144}\ m_1\ m_3\ g_1^2\ g_3^2\ LF_{1,1,1,1,0}[m_1,\ m_3,\ m_{\tilde{q}}^{i3},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3} - \\
&\frac{1}{32}\ g_2^2\ g_3^2\ LF_{1,1,1,1,-1}[m_2,\ m_3,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3} - \\
&\frac{1}{16}\ m_2\ m_3\ g_2^2\ g_3^2\ LF_{1,1,1,1,0}[m_2,\ m_3,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3} - \\
&\frac{1}{48}\ g_2^2\ g_3^2\ LF_{1,1,1,1,-1}[m_2,\ m_3,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4} - \\
&\frac{1}{24}\ m_2\ m_3\ g_2^2\ g_3^2\ LF_{1,1,1,1,0}[m_2,\ m_3,\ m_{\tilde{q}}^{i1},\ m_{\tilde{q}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4} - \\
&\frac{1}{48}\ g_2^2\ g_3^2\ LF_{1,1,1,1,-1}[m_2,\ m_3,\ m_{\tilde{q}}^{i2},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4} - \\
&\frac{1}{24}\ m_2\ m_3\ g_2^2\ g_3^2\ LF_{1,1,1,1,0}[m_2,\ m_3,\ m_{\tilde{q}}^{i2},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4} - \\
&\frac{1}{32}\ g_2^2\ g_3^2\ LF_{1,1,1,1,-1}[m_2,\ m_3,\ m_{\tilde{q}}^{i3},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3} - \\
&\frac{1}{16}\ m_2\ m_3\ g_2^2\ g_3^2\ LF_{1,1,1,1,0}[m_2,\ m_3,\ m_{\tilde{q}}^{i3},\ m_{\tilde{q}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3})
\end{aligned}
$$